TapRunner User Manual

Sep 28, 2026 · @Robert LaMoreaux

Overview

TapRunner is a Windows desktop app for running OpenTAP TapPlans from a test station. Instead of using OpenTAP's own command line or Editor directly, an operator opens a plan, fills in a few parameters (like a serial number), and presses Run - TapRunner launches `tap.exe`, shows six color status circles that update live as the plan runs, and keeps a written log of everything that happened, including a separate log of exactly which steps failed.

It's built for repeat use on a fixed test station: the last plan used reopens automatically, default parameter values can be saved once and reused, and a runtime clock shows how long each run is taking.

Installing TapRunner

1. Copy `TapRunnerSetup.msi` to the station and double-click it (or run `msiexec /i TapRunnerSetup.msi` for a silent install). No administrator rights or UAC prompt are needed - it installs for the current Windows user only, under `%LocalAppData%\Programs\TapRunner`, and adds a Start Menu shortcut.
2. The installer also drops the companion OpenTAP plugin (`TapRunner.StatusListener`) and, if `tap.exe` is found on the system `PATH`, installs it into OpenTAP automatically. From version 1.3.5 the installer runs that install from tap.exe's own folder; earlier installers could fail silently on stations whose instrument plugins read config files from the current folder (such as the Riker switch matrix's RSMDEFINE.CSV). To check which plugin OpenTAP has, look at the Version in C:\OpenTAP\Packages\TapRunner.StatusListener\package.xml.
3. **One manual step remains, and only needs doing once per station:** open the OpenTAP Editor or TUI, go to *Settings* → *Results*, and make sure **TapRunner Status** is added and enabled as a result listener alongside whatever other listeners are already configured. Without this step TapRunner still runs plans, but the status circles won't update live.

To remove TapRunner, use Windows Settings → Apps, or run the same installer again. Uninstalling does not remove the plugin from OpenTAP.

First-Time Setup

Click **Settings...** in the main window to open the Settings dialog. Most fields only need to be checked once per station.

Field	What it's for
OpenTAP executable (tap.exe)	Full path to `tap.exe`. Defaults to `C:\OpenTAP\tap.exe`.
OpenTAP profile	Which bench settings profile to run with, picked from a dropdown of the profiles found under

Field	What it's for
	`Settings\Bench` next to `tap.exe` (or typed in directly). Leave blank to use OpenTAP's own default.
SQLite results database	The database file OpenTAP's SQLite result listener stores test results in. TapRunner reads it from, and saves it back to, OpenTAP's own Settings\Results.xml, so the change applies to every run from then on. Browse to an existing .TapResults file to add to it, or type a new name to start one. The previous Results.xml is kept as Results.xml.bak. If the field is greyed out, add a SQLite result listener in OpenTAP first. Close the OpenTAP Editor before changing it, or it may save over the change.
Test results viewer	The program the main window's View Results button opens the SQLite results database in. Defaults to C:\OpenTAP\TestRunViewer\TestRunViewerSqlite.exe.
Working directory	The folder `tap.exe` is launched from.
Station name	Shown next to Operator in the main window. Defaults to the Windows computer name the first time TapRunner ever runs.
TapRunner log file / OpenTAP log file	The TapRunner log file is where TapRunner's own running log is written (every status message and completed step, timestamped, kept separate from OpenTAP's log); the failure log is saved next to it automatically. The Open TapRunner Log button next to it opens it. The OpenTAP log file is the one the main window's Open Log button opens; leave it blank to use SessionLogs\Latest.txt next to tap.exe.
Clear the failure log at the start of each run	Off by default (the failure log keeps accumulating across runs, each headed with the plan name and start time). Turn on to have it hold only the current run's failures.
Default parameter values	On the Parameters tab: a saved list of parameter name → value overrides that apply every time a matching plan loads - see "Running a Test" below. **Copy From Current Plan** fills this list from whatever's currently shown in the main window.

Field	What it's for
	Tick Hidden on a row to keep that parameter out of the main window's list - it is still passed to the test with its default value (or the plan's own if left blank); a checkbox under the list shows hidden parameters greyed out (still editable) instead.
Status colors	The color (and whether it flashes) for each of the six status circles.
Power On plan / Power Off plan	The plans run by the small Power On and Power Off buttons on the main window. Default PowerON.TapPlan and PowerOFF.TapPlan; a bare name is looked for in the working directory, then next to tap.exe, or browse to a full path.

Click **OK** to save changes, or **Cancel** to discard them.

Running a Test

1. **Open a plan.** Click **Browse...** and pick a `.TapPlan` file, or let TapRunner reopen the last one automatically on startup. Its external (test plan) parameters appear in a list, sorted alphabetically.
2. **Check the parameters.** Any parameter with "serial" in its name gets keyboard focus automatically, with its value selected, so a barcode scan overwrites it in one motion. Other parameters are pre-filled from a saved default (Settings → Default parameter values) if one exists, otherwise from the plan's own default.
3. **Load Defaults**, next to the parameter list, re-applies the saved defaults to what's currently shown - useful after changing them in Settings without reopening the plan. Anything typed by hand afterward is never overwritten automatically.
4. **Choose Run or Edit.** With "Edit with TUI" unchecked, the button reads **Run** and launches the plan headless, passing along every parameter value. Checked, it reads **Edit** and opens OpenTAP's text-based interface instead - useful for stepping through or adjusting a plan by hand, though parameter values and the selected profile aren't passed through to TUI mode.
5. **Watch it run.** TapRunner brings itself back to the front a moment after launching (the console window can otherwise steal focus), and the status circles, step history, and runtime clock update live - see "Status Indicators" below.

Power On and Power Off. The small **Power On** and **Power Off** buttons next to Settings run the plans chosen under Settings (`PowerON.TapPlan` and `PowerOFF.TapPlan` by default). They always run headless, whatever the "Edit with TUI" box says, through the same status circles, runtime clock and logs as a normal run. They use the default parameter values from Settings and ignore whatever is typed in the main window's parameter list, and they're disabled while any run is in progress.

Status Indicators

Six circles on the right show the run's state, each lighting in its configured color (Settings → Status colors):

Circle	Lights when
Ready	Idle - no test currently running
Running	A plan is executing (flashes by default)
Passed	The run finished with a Pass verdict; also flashes while running for as long as the test is passing
Failed	The run finished with a Fail verdict; also flashes while running when a step fails. If a parent step then clears that failure (e.g. a retry that passes), it goes dim instead and Passed flashes again - the dim circle stays after the run to show the pass followed a resolved failure
Error	The run finished with an Error verdict, including if the process ended without ever reporting a result (never guessed as Ready); flashes and dims while running the same way as Failed
Aborted	The run was stopped via Abort, or OpenTAP's own cooperative abort finished it

Below the circles and the logo sits a **runtime clock** (`hh:mm:ss`) that starts the moment `tap.exe` is launched and freezes the instant the run ends, so it's easy to see how long a run is taking or took.

The status line above the buttons shows "Running: `<step name>`..." as each step starts, even for steps that never publish a result. The scrollable box below the buttons keeps a running history of every completed step (name, verdict, any published results, and the error message if the step threw), up to 1000 lines, growing from the top. Making the window taller makes this box taller too, so more history stays in view.

Stopping and Aborting a Run

Button	What it does
Stop	Asks OpenTAP to abort gracefully - it finishes the current step and runs any cleanup, then ends (usually with an Aborted verdict). Only works if TapRunner Status is enabled in OpenTAP's Results settings; otherwise use Abort.
Abort	Immediately kills the `tap.exe` process, with a confirmation prompt first. Use this if Stop isn't available or a run needs to end right away.

Closing TapRunner while a test is running also asks for confirmation, since doing so aborts the run.

Logs

Button	Opens

Button	Opens
Open Log	OpenTAP's own live log (not TapRunner's), in whatever program the station uses for `.txt` files. TapRunner's own log is opened from Settings instead (Open TapRunner Log).
Failed	A separate TapRunner window listing every step that finished Fail or Error, with fuller detail than the main step history - the plan name and start time head each run's entries. Click **Refresh** to pick up new failures while a run is still in progress. Each failed measurement step has a Limits line - its minimum and maximum limits - above the value it measured (see "Limits in the failure log" below), and Save As... saves a copy of the whole failure log to a file you choose.
View Results	The SQLite results database your test results are stored in, opened in the test results viewer set under Settings.

By default the failure log keeps accumulating across runs. Check **Settings → Clear the failure log at the start of each run** to have it hold only the most recent run's failures instead.

An errored step's entry includes the error message it threw. If a run ends in an Error that no step reported - for example an instrument failed to open before any step ran, or TapRunner never got a result back - the failure log gets a "(test plan)" entry instead, so an Error run never leaves it empty. That entry (also shown in the history box and the status line) includes whatever OpenTAP itself logged as an error during the run - for example a failed licence check - taken from its session log.

Limits in the failure log. Each failed step's entry shows a Limits line with the step's own limit settings, as OpenTAP recorded them for that run: its "Minimum Value" and "Maximum Value" become Min= and Max=, any other setting in the step's Limits group is listed as Name=Value, and "(limit check off)" is added when the step's "Check Limits" setting is off. Steps with no limit settings, such as group steps like Check VREFs, have no Limits line. This needs TapRunner 1.3.5 or later, with its TapRunner Status plugin installed in OpenTAP. An entry from PowerON.TapPlan:

```
[2026-10-28 11:21:32] FAIL  Read QCA Voltage ON
----------------------------------------------------------------------
Limits: Min=3.2, Max=3.4
Read Measurement: Measurement=AIN_V_3V3QCA
Value=1.99707
```

Troubleshooting

Symptom	Likely cause / fix
Status circles never move past Ready/Running	**TapRunner Status** isn't enabled as a result listener in OpenTAP (Settings → Results). Add/enable it - see "Installing TapRunner".
"The configured OpenTAP executable was not found"	Fix the path under Settings → OpenTAP executable (tap.exe).

Symptom	Likely cause / fix
Run finishes with Error and no clear reason	The process likely exited without reporting a result - a licensing error or startup failure is the most common cause. The status line, history box and failure log now show the error OpenTAP logged (such as the licence message); the Failed button has the full entry.
Parameter values or the OpenTAP profile don't apply	Both are only passed through in headless Run mode - TUI ("Edit") mode doesn't accept either, by OpenTAP's own design.
A saved default parameter value doesn't show up	Check it's saved under the parameter's short name (e.g. `PowerOn`), not OpenTAP's full internal name (`Parameters \ PowerOn`).
Stop button has no effect	No status listener is currently connected for this run - use Abort instead.
A parameter is missing from the list	It is probably marked Hidden in Settings (Default parameter values, Hidden column). It is still passed to the test. Untick Hidden to show it, or tick "Show hidden parameters greyed out".
Power On / Power Off says the plan was not found	Check the plan names under Settings. A bare name is looked for in the working directory, then next to tap.exe; or browse to the plan's full path.
A failed step has no Limits line in the failure log	Either the step has no limit settings (group steps never do), or OpenTAP still has a TapRunner Status plugin older than 1.3.5. Check the Version in C:\OpenTAP\Packages\TapRunner.StatusListener\package.xml, and reinstall with TapRunnerSetup.msi if it is older. To install the plugin by hand, run tap package install from the C:\OpenTAP folder, not from anywhere else.

Still stuck? Check the logs first - TapRunner's (Settings > Open TapRunner Log) and OpenTAP's (**Open Log**) - together they record every status change and OpenTAP message during the run, and are usually the fastest way to see what happened.